

María Esther Hernández Ortega

Robotics and Mechatronics Engineer, Zacatecas, Zac. · · +52 492 161 8803 · · mariestherhernandezortega@gmail.com

Robotics and Mechatronics Engineering student with a strong background in embedded systems development, device control using microcontrollers, and process automation. Recognized for responsibility, teamwork, effective communication, and commitment to continuous improvement. Passionate about technological innovation and focused on the practical application of acquired knowledge to develop efficient and high-impact solutions.

PROFESSIONAL EXPERIENCE

Labsol Network (Council of Science, Technology and Innovation of Zacatecas)
Social Service

Zacatecas, Zacatecas
October 2024–May 2025

- Development of interactive systems for the reuse of stationary bicycles, promoting physical activity in institutional environments.
- Design and implementation of an LED race system controlled by spinning bikes, integrating Arduino microcontrollers, sensors, and actuators, using object-oriented programming in C++.
- Creation and optimization of code to improve system performance. Time management and task planning to ensure timely and functional project delivery.

VEIMAZ (Agricultural Machinery Workshop)
Full Stack Developer – Engineering and Automation

Col. Hidalgo, Sombrerete, Zacatecas
May 2025– August 2025

- Operation and control of a CNC plasma cutting machine, as well as design of parts and cutting paths in AutoCAD for the manufacturing of agricultural implements.

EDUCATION

AUTONOMOUS UNIVERSITY OF ZACATECAS
Robotics and Mechatronics Engineering

Zacatecas, Zacatecas
August 2021-Currently

ASOCIACIÓN DE ROBOTICA APLICADA Y CIENCIAS DE LA TECNOLOGIA
Advanced Course in Applied Robotics – Certified by the STPS
Honors: Diploma of Excellence
Python Course -- Certified by the STPS
Honors: Diploma of Excellence

AUTONOMOUS UNIVERSITY OF ZACATECAS
Introduction to Desktop Development with Visual Studio using the .NET Framework Course

GOOGLE COURSERA
Google AI Essentials Course

ADDITIONAL SKILLS

• Python	• Solidwork	• C, C++	• Microsoft Office	• Edge Impulse
• Micro Python	• AutoCAD	• HTML	• Microsoft Azure	• Altium Designer
• Verilog(FPGA)	• Proteus	• Visual Studio	• Sensores/Actuadores	• Node Red

LANGUAGES

- Español – Native
- Inglés – Intermediate

REFERENCES

- ITIC. Alfonso Rafael Solis Rangel +52 458 126 3350
- Eng. Marco Hernández González +524331083997
Owner VEIMAZ